

CS 2601 Linear and Convex Optimization

3. Convex sets

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Fall 2023

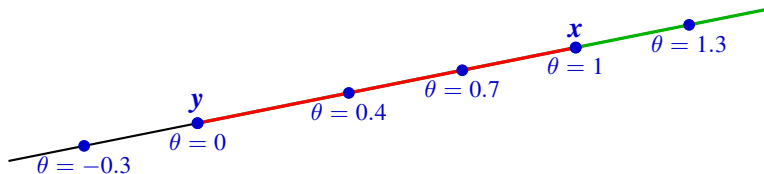
Outline

- Convex Sets
- Supporting and separating hyperplanes

Lines, line segments and rays

Given $x \neq y \in \mathbb{R}^n$, the **line** passing through x and y consists of points of the form

$$z = y + \theta(x - y) = \theta x + (1 - \theta)y, \quad \theta \in \mathbb{R}$$



The **ray (half-line)** with endpoint y and direction $x - y$ consists of points

$$\theta x + (1 - \theta)y, \quad \theta \geq 0$$

The **line segment** between x and y consists of points

$$\theta x + (1 - \theta)y, \quad 0 \leq \theta \leq 1$$

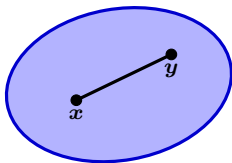
Note. Often use notation $\bar{\theta} = 1 - \theta$.

Convex sets

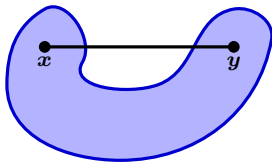
A set $C \subset \mathbb{R}^n$ is **convex** if the line segment between any two points $x, y \in C$ lies entirely in C , i.e.

$$x \in C, y \in C, \theta \in [0, 1] \implies \theta x + \bar{\theta} y \in C$$

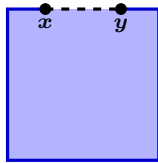
Note. Only need to check the case that $x \neq y$ and $\theta \in (0, 1)$.



convex



nonconvex



nonconvex

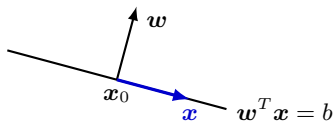
For $\theta \in [0, 1]$, $\theta x + \bar{\theta} y$ is called a **convex combination** of x and y . In a more symmetric form, a convex combination is

$$\theta_1 x + \theta_2 y \quad \text{where } \theta_1 \geq 0, \theta_2 \geq 0, \theta_1 + \theta_2 = 1$$

Examples of convex sets

Example. Trivial examples of convex sets include empty set, $\mathbb{R}^n$, singletons (points), lines, line segments and rays.

Example. A hyperplane $P = \{x \in \mathbb{R}^n : w^T x = b\}$ is convex, where $w \in \mathbb{R}^n$, $b \in \mathbb{R}$.

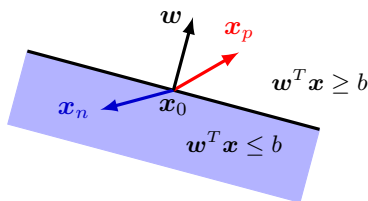


Proof. For $x_1, x_2 \in P$ and $\theta \in [0, 1]$,

$$\begin{aligned} w^T(\theta x_1 + \bar{\theta} x_2) &= \theta w^T x_1 + \bar{\theta} w^T x_2 \\ &= \theta b + \bar{\theta} b = b \end{aligned}$$

Example: Halfspaces

A halfspace $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{w}^T \mathbf{x} \leq b\}$ is convex.



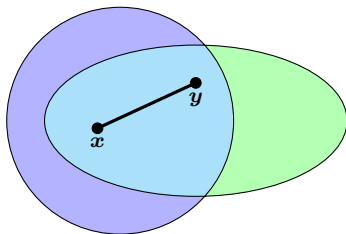
Note. $H = \{\mathbf{x} : f(\mathbf{x}) \leq b\}$ is the so-called sublevel set of $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$. Note $\nabla f(\mathbf{x}) = \mathbf{w}$, the outward normal to the boundary hyperplane.

Proof. For $\mathbf{x}_1, \mathbf{x}_2 \in H$ and $\theta \in [0, 1]$,

$$\begin{aligned}\mathbf{w}^T(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) &= \theta \mathbf{w}^T \mathbf{x}_1 + \bar{\theta} \mathbf{w}^T \mathbf{x}_2 \\ &\leq \theta b + \bar{\theta} b = b\end{aligned}$$

Set intersection preserves convexity

Proposition. The intersection of an arbitrary collection of convex sets is convex.



Proof. Let $\{C_i : i \in I\}$ be an arbitrary collection of convex sets with index set I , and $C = \bigcap_{i \in I} C_i$ their intersection.

- Let $x, y \in C$, $\theta \in [0, 1]$
- $x, y \in C_i$ for any $i \in I$
- By convexity of C_i , $\theta x + \bar{\theta} y \in C_i$
- $\theta x + \bar{\theta} y \in C$

Example: Affine spaces

An affine space $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{b}\}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ is convex.

Note. An affine space is a shifted linear space, $S = \mathbf{x}_0 + S_0$, where $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$, and $S_0 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}\}$ is a linear space.

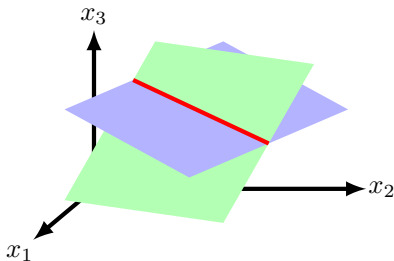
Can verify convexity by definition; here use the intersection property.

- let $\mathbf{A}^T = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m]$,
 $\mathbf{b} = (b_1, \dots, b_m)$
- S is intersection of m hyperplanes

$$S = \bigcap_{i=1}^m P_i$$

where

$$P_i = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i\}$$



Note. An affine space S actually contains the line through any $\mathbf{x}, \mathbf{y} \in S$. 7

Example: Polyhedra

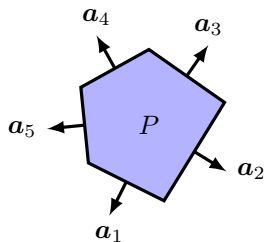
A polyhedron $P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} \leq \mathbf{b}\}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ is convex, where vector inequality $\leq$ is interpreted componentwise

- let $\mathbf{A}^T = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m]$, $\mathbf{b} = (b_1, \dots, b_m)$
- P is intersection of m halfspaces

$$P = \bigcap_{i=1}^m H_i$$

where

$$H_i = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} \leq b_i\}$$



Note. An affine space $S = \{\mathbf{x} : \mathbf{Ax} = \mathbf{b}\}$ is a polyhedron

$$\mathbf{Ax} = \mathbf{b} \iff (\mathbf{Ax} \leq \mathbf{b} \text{ and } -\mathbf{Ax} \leq -\mathbf{b}) \iff \begin{bmatrix} \mathbf{A} \\ -\mathbf{A} \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \end{bmatrix}$$

More generally, $\{\mathbf{x} : \mathbf{Ax} \leq \mathbf{b}, \mathbf{Cx} = \mathbf{d}\}$ is a polyhedron.

Example: Polyhedra (cont'd)

The set of points satisfying the following inequalities is a polyhedron

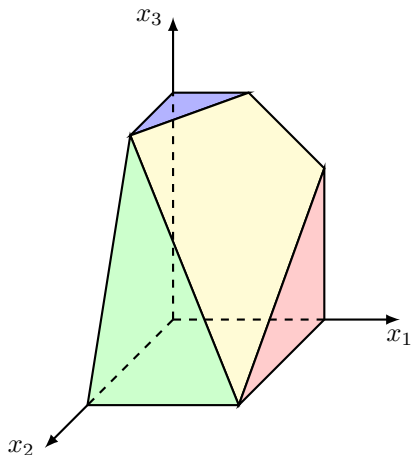
$$x_1 + x_2 + x_3 \leq 4$$

$$x_1 \leq 2$$

$$x_3 \leq 3$$

$$3x_2 + x_3 \leq 6$$

$$\mathbf{x} \geq \mathbf{0}$$



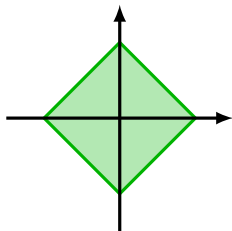
Example: Polyhedra (cont'd)

The 1-norm unit ball is a polyhedron,

$$B_1 = \{\mathbf{x} : \|\mathbf{x}\|_1 \leq 1\}$$

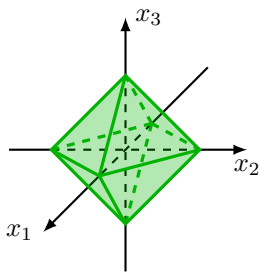
In 2D,

$$B_1^{(2)} = \{\mathbf{x} : \begin{aligned} x_1 + x_2 &\leq 1, \\ x_1 - x_2 &\leq 1, \\ -x_1 + x_2 &\leq 1, \\ -x_1 - x_2 &\leq 1 \end{aligned}\}$$



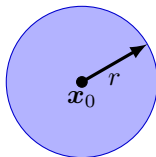
In 3D,

$$B_1^{(3)} = \{\mathbf{x} : \pm x_1 \pm x_2 \pm x_3 \leq 1\}$$



Example: Norm balls

A closed ball $\bar{B}(\mathbf{x}_0, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$ is convex.



Proof. For $\mathbf{x}_1, \mathbf{x}_2 \in \bar{B}(\mathbf{x}_0, r)$ and $\theta \in [0, 1]$,

$$\begin{aligned}\|(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) - \mathbf{x}_0\| &= \|\theta(\mathbf{x}_1 - \mathbf{x}_0) + \bar{\theta}(\mathbf{x}_2 - \mathbf{x}_0)\| \\ &\leq \|\theta(\mathbf{x}_1 - \mathbf{x}_0)\| + \|\bar{\theta}(\mathbf{x}_2 - \mathbf{x}_0)\| \\ &= \theta \|\mathbf{x}_1 - \mathbf{x}_0\| + \bar{\theta} \|\mathbf{x}_2 - \mathbf{x}_0\| \leq r\end{aligned}$$

Note. True for any norm $\|\cdot\|$.

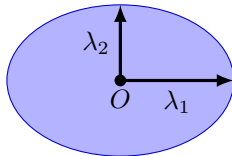
Note. Open balls are also convex.

Example: Ellipsoids

An ellipsoid

$$\mathcal{E} = \left\{ \mathbf{x} \in \mathbb{R}^2 : \frac{x_1^2}{\lambda_1^2} + \frac{x_2^2}{\lambda_2^2} \leq 1 \right\}$$

is convex.



Proof. Let $\mathbf{\Lambda} = \text{diag}\{\lambda_1, \lambda_2\}$. Note

$$\mathcal{E} = \{\mathbf{x} : \mathbf{x}^T \mathbf{\Lambda}^{-2} \mathbf{x} \leq 1\} = \{\mathbf{x} : \|\mathbf{\Lambda}^{-1} \mathbf{x}\|_2 \leq 1\} = \{\mathbf{\Lambda} \mathbf{u} : \|\mathbf{u}\|_2 \leq 1\}.$$

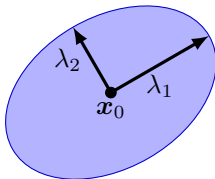
For $\mathbf{x}_i = \mathbf{\Lambda} \mathbf{u}_i \in \mathcal{E}$, and $\theta \in [0, 1]$,

$$\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2 = \mathbf{\Lambda}(\theta \mathbf{u}_1 + \bar{\theta} \mathbf{u}_2).$$

Recall the unit ball is convex, so $\|\theta \mathbf{u}_1 + \bar{\theta} \mathbf{u}_2\|_2 \leq 1$ and $\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2 \in \mathcal{E}$. 12

Example: Ellipsoids (cont'd)

An ellipsoid $\mathcal{E} = \{\mathbf{x}_0 + \mathbf{A}\mathbf{u} : \|\mathbf{u}\|_2 \leq 1\}$, $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{A} \succ \mathbf{0}$ is convex.



$\mathbf{A}$ has eigendecomposition $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$, where $\mathbf{\Lambda}$ is diagonal and $\mathbf{Q}$ is orthogonal. With $\tilde{\mathbf{u}} = \mathbf{Q}^T\mathbf{u}$,

$$\mathcal{E} = \{\mathbf{x}_0 + \mathbf{Q}\mathbf{\Lambda}\tilde{\mathbf{u}} : \|\tilde{\mathbf{u}}\|_2 \leq 1\},$$

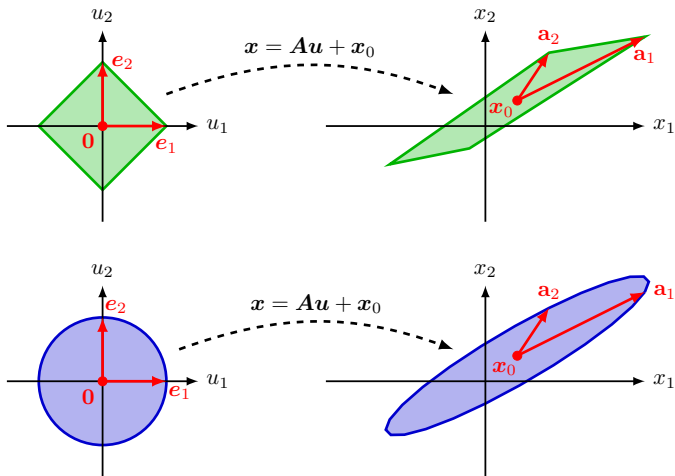
which is a rotated and shifted version of $\mathcal{E}' = \{\mathbf{\Lambda}\tilde{\mathbf{u}} : \|\tilde{\mathbf{u}}\|_2 \leq 1\}$.

Note. The lengths of semi-axes are eigenvalues of $\mathbf{A}$

Note. Also often written as $\mathcal{E} = \{\mathbf{x} : (\mathbf{x} - \mathbf{x}_0)^T \mathbf{P}^{-1}(\mathbf{x} - \mathbf{x}_0) \leq 1\}$, $\mathbf{P} = \mathbf{A}^2$.

Affine transformation preserves convexity

$$\mathbf{x} = \mathbf{A}\mathbf{u} + \mathbf{x}_0, \quad \text{where } \mathbf{A} = \begin{bmatrix} 2 & 0.5 \\ 1 & 0.75 \end{bmatrix}, \quad \mathbf{x}_0 = (0.5, 0.4)$$



Affine transformation preserves convexity

Proposition. The image of a convex set under an affine transformation is convex.

Proof. Let $C \subset \mathbb{R}^n$ be a convex set and $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ an affine transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Given $\mathbf{y}_1, \mathbf{y}_2 \in f(C) = \{f(\mathbf{x}) : \mathbf{x} \in C\}$ and $\theta \in [0, 1]$, need to show $\theta\mathbf{y}_1 + \bar{\theta}\mathbf{y}_2 \in f(C)$.

1. By definition, $\mathbf{y}_i = f(\mathbf{x}_i)$ for some $\mathbf{x}_i \in C$, $i = 1, 2$.
2. Since f is affine,

$$\begin{aligned}\theta\mathbf{y}_1 + \bar{\theta}\mathbf{y}_2 &= \theta f(\mathbf{x}_1) + \bar{\theta} f(\mathbf{x}_2) \\ &= \theta(\mathbf{A}\mathbf{x}_1 + \mathbf{b}) + \bar{\theta}(\mathbf{A}\mathbf{x}_2 + \mathbf{b}) \\ &= \mathbf{A}(\theta\mathbf{x}_1 + \bar{\theta}\mathbf{x}_2) + \mathbf{b}\end{aligned}$$

3. Since C is convex, $\mathbf{z} \triangleq \theta\mathbf{x}_1 + \bar{\theta}\mathbf{x}_2 \in C$, so $\theta\mathbf{y}_1 + \bar{\theta}\mathbf{y}_2 = f(\mathbf{z}) \in f(C)$.

Proposition. The inverse image of a convex set under an affine transformation is convex.

Example: Positive semidefinite matrices

The set of positive semidefinite matrices

$$\mathcal{S}_+^n = \{\mathbf{A} \in \mathbb{R}^{n \times n} : \mathbf{A} \succeq \mathbf{O}\}$$

is convex.

Proof. For arbitrary $\mathbf{A}, \mathbf{B} \in \mathcal{S}_+^n$ and $\theta \in [0, 1]$, $\mathbf{x} \in \mathbb{R}^n$, need to show $\theta\mathbf{A} + \bar{\theta}\mathbf{B} \in \mathcal{S}_+^n$. Check the definition of positive semidefiniteness.

1. $\theta\mathbf{A} + \bar{\theta}\mathbf{B}$ is symmetric,

$$(\theta\mathbf{A} + \bar{\theta}\mathbf{B})^T = \theta\mathbf{A}^T + \bar{\theta}\mathbf{B}^T = \theta\mathbf{A} + \bar{\theta}\mathbf{B}$$

2. $\mathbf{x}^T(\theta\mathbf{A} + \bar{\theta}\mathbf{B})\mathbf{x} \geq 0$ for any $\mathbf{x}$,

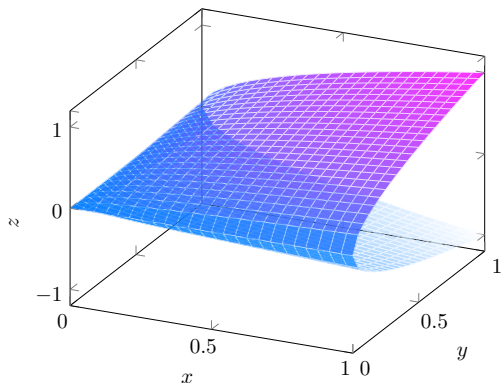
$$\mathbf{x}^T(\theta\mathbf{A} + \bar{\theta}\mathbf{B})\mathbf{x} = \theta(\mathbf{x}^T\mathbf{A}\mathbf{x}) + \bar{\theta}(\mathbf{x}^T\mathbf{B}\mathbf{x}) \geq 0$$

Example: Positive semidefinite matrices (cont'd)

For $n = 2$, can identify $\mathcal{S}_+^2$ with a subset of $\mathbb{R}^3$. By Sylvester's Theorem,

$$\mathbf{A} = \begin{bmatrix} x & z \\ z & y \end{bmatrix} \in \mathcal{S}_+^2 \iff x \geq 0, y \geq 0, xy \geq z^2$$

Boundary $\partial\mathcal{S}_+^2 = \{(x, y, z) : x \geq 0, y \geq 0, z^2 = xy\}$



Convex combination

A **convex combination** of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in \mathbb{R}^n$ is a point of the form

$$\sum_{i=1}^m \theta_i \mathbf{x}_i = \theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \dots + \theta_m \mathbf{x}_m$$

where $\theta_i \geq 0$ for all i and $\sum_{i=1}^m \theta_i = 1$.

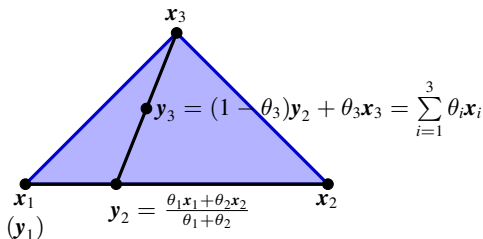
Theorem. If C is convex, and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in C$, then any convex combination $\sum_{i=1}^m \theta_i \mathbf{x}_i \in C$.

In general, $\mathbf{y}_1 = \mathbf{x}_1$, and

$$\mathbf{y}_k = \frac{\sigma_{k-1}}{\sigma_k} \mathbf{y}_{k-1} + \frac{\theta_k}{\sigma_k} \mathbf{x}_k, \quad k \geq 2$$

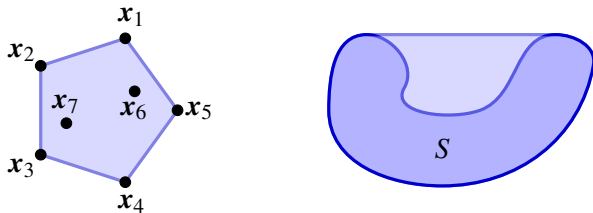
where

$$\sigma_k = \sum_{i=1}^k \theta_i$$



Convex hull

The **convex hull** of a set $S \subset \mathbb{R}^n$, denoted $\text{conv } S$, is the smallest convex set containing S .



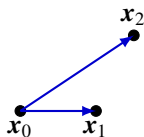
Theorem. $\text{conv } S$ is the set of all convex combinations of points in S , i.e.

$$\text{conv } S = \left\{ \sum_{i=1}^m \theta_i \mathbf{x}_i : m \in \mathbb{N}; \mathbf{x}_i \in S, \theta_i \geq 0, i = 1, \dots, m; \sum_{i=1}^m \theta_i = 1 \right\}$$

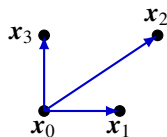
Note. Actually we can replace $m \in \mathbb{N}$ by $m \leq n + 1$ in the above representation, i.e. each $\mathbf{x} \in \text{conv } S$ is the convex combination of at most $n + 1$ points in S .

Affinely independent points

$m + 1$ points $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$ are **affinely independent** if $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_m - \mathbf{x}_0$ are linearly independent.



affinely independent points in $\mathbb{R}^2$



affinely dependent points in $\mathbb{R}^2$

Proposition. $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$ are affinely independent iff

$$\sum_{i=0}^m c_i \mathbf{x}_i = \mathbf{0} \text{ and } \sum_{i=0}^m c_i = 0 \implies c_i = 0 \text{ for } i = 0, 1, \dots, m$$

Note. In $\mathbb{R}^n$, the maximum number of linearly independent vectors is n , so the maximum number of affinely independent points is $n + 1$.

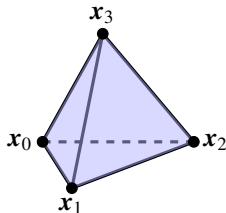
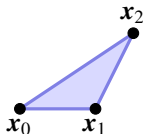
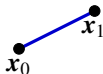
Simplexes

An m -dimensional **simplex**, also called an **m -simplex**, is the convex hull of $m + 1$ affinely independent points. More specifically, the simplex determined by affinely independent points $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m$ is

$$\text{conv}\{\mathbf{x}_0, \dots, \mathbf{x}_m\} = \{\theta_0 \mathbf{x}_0 + \theta_1 \mathbf{x}_1 + \dots + \theta_m \mathbf{x}_m : \theta \geq \mathbf{0}, \mathbf{1}^T \theta = 1\}$$

Note. $\mathbb{R}^n$ only has m -simplexes with $m \leq n$

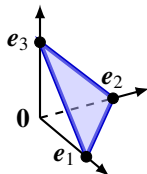
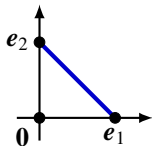
- 0-simplexes are points
- 1-simplexes are line segments
- 2-simplexes are triangles
- 3-simplexes are tetrahedra



Simplexes (cont'd)

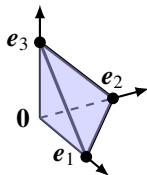
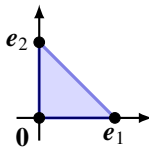
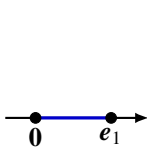
Example. The **probability n -simplex** is the n -simplex in $\mathbb{R}^{n+1}$ determined by the standard basis vectors $e_1, \dots, e_{n+1}$,

$$\Delta_n = \{\theta \in \mathbb{R}^{n+1} : \theta \geq 0, \mathbf{1}^T \theta = 1\}$$



Example. The **unit n -simplex** in $\mathbb{R}^n$ is the n -simplex determined by $0 \in \mathbb{R}^n$ and the standard basis vectors $e_1, \dots, e_n \in \mathbb{R}^n$,

$$\Delta'_n = \{\theta' \in \mathbb{R}^n : \theta' \geq 0, \mathbf{1}^T \theta' \leq 1\}$$



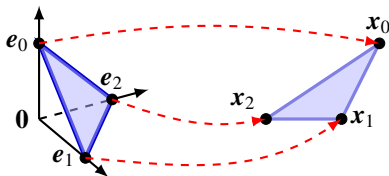
Simplexes (cont'd)

The m -simplex in $\mathbb{R}^n$ determined by affinely independent points $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m$ is the image of Δ_m under a linear transformation

$$\boldsymbol{\theta} = \sum_{i=0}^m \theta_i \mathbf{e}_i \mapsto \mathbf{x} = \sum_{i=0}^m \theta_i \mathbf{x}_i = \mathbf{X}\boldsymbol{\theta}$$

where

$$\mathbf{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m] \in \mathbb{R}^{n \times (m+1)}, \quad \boldsymbol{\theta} = (\theta_0, \theta_1, \dots, \theta_m) \in \Delta_m$$



Simplexes (cont'd)

Note

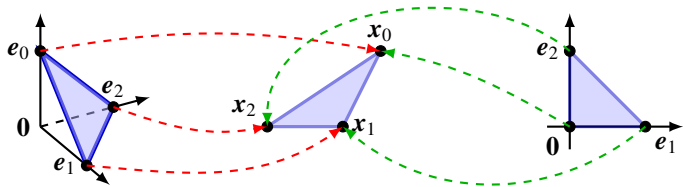
$$\mathbf{x} = \sum_{i=0}^m \theta_i \mathbf{x}_i = \mathbf{x}_0 + \sum_{i=1}^m \theta_i (\mathbf{x}_i - \mathbf{x}_0)$$

and $\boldsymbol{\theta}' = (\theta_1, \dots, \theta_m) \in \Delta'_m$.

The simplex $\text{conv}\{\mathbf{x}_0, \dots, \mathbf{x}_m\}$ is also the image of Δ'_m under the affine transformation

$$\boldsymbol{\theta}' \mapsto \mathbf{x} = \mathbf{x}_0 + \mathbf{B}\boldsymbol{\theta}'$$

where $\mathbf{B} = [\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_m - \mathbf{x}_0] \in \mathbb{R}^{n \times m}$.

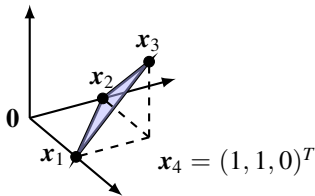


Simplexes (cont'd)

Example. Let $\mathbf{x}_1 = (1, 0, 0)$, $\mathbf{x}_2 = (0, 1, 0)$ and $\mathbf{x}_3 = (1, 1, 1)$.
Points in the 2-simplex determined by $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are of the form

$$\mathbf{x} = \sum_{i=1}^3 \theta_i \mathbf{x}_i = (\theta_1 + \theta_3, \theta_2 + \theta_3, \theta_3)$$

where $\boldsymbol{\theta} \in \Delta_2$, i.e. $\theta_i \geq 0$, $\theta_1 + \theta_2 + \theta_3 = 1$.



Alternatively,

$$\mathbf{x} = \mathbf{x}_1 + \theta_2(\mathbf{x}_2 - \mathbf{x}_1) + \theta_3(\mathbf{x}_3 - \mathbf{x}_1) = (1 - \theta_2, \theta_2 + \theta_3, \theta_3) = \mathbf{x}_1 + \mathbf{B}\boldsymbol{\theta}',$$

where

$$\mathbf{B} = [\mathbf{x}_2 - \mathbf{x}_1, \mathbf{x}_3 - \mathbf{x}_1] = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\theta}' = (\theta_2, \theta_3) \in \Delta'_2$$

Note $\mathbf{B}$ has full column rank by the affine independence of the $\mathbf{x}_i$'s.

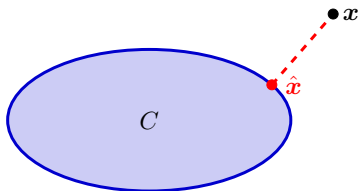
Outline

- Convex Sets
- Supporting and separating hyperplanes

Projection onto convex set

Given a set $C \subset \mathbb{R}^n$, the distance between a point x and C is

$$\text{dist}(\mathbf{x}, C) = \inf_{z \in C} \|\mathbf{x} - z\|$$



Theorem. If $C \subset \mathbb{R}^n$ is nonempty, closed and convex, then for any x , there is a unique $\hat{x} \in C$ s.t.

$$\text{dist}(\mathbf{x}, C) = \|\mathbf{x} - \hat{\mathbf{x}}\|$$

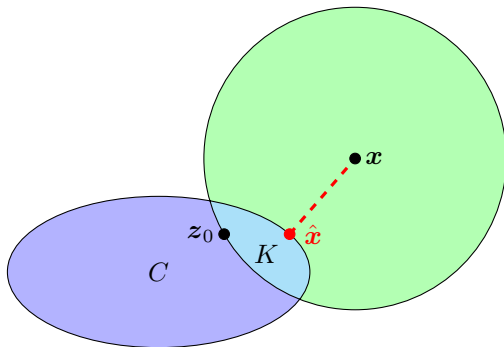
$\hat{x}$ is called the **projection** of x onto C and denoted by $\hat{x} = \mathcal{P}_C(x)$.

Note. $\mathcal{P}_C(\mathbf{x}) = \mathbf{x}$ iff $\mathbf{x} \in C$.

Projection onto convex set (cont'd)

Proof. First show existence.

- Let $z_0 \in C$. Then $\text{dist}(\mathbf{x}, C) \leq \|\mathbf{x} - z_0\|$.
- Let $K = \{z \in C : \|\mathbf{x} - z\| \leq \|\mathbf{x} - z_0\|\} = C \cap \bar{B}(\mathbf{x}, \|\mathbf{x} - z_0\|)$.
- $\|\mathbf{x} - z\|$ is continuous in z , K compact $\implies \exists \hat{\mathbf{x}} \in K$ such that $\text{dist}(\mathbf{x}, C) = \|\mathbf{x} - \hat{\mathbf{x}}\|$



Projection onto convex set (cont'd)

Proof (cont'd). Now show uniqueness. Suppose $\hat{x}_1, \hat{x}_2 \in C$ satisfy $\text{dist}(\mathbf{x}, C) = \|\mathbf{x} - \hat{x}_1\| = \|\mathbf{x} - \hat{x}_2\|$.

- $\hat{x}_c := \frac{\hat{x}_1 + \hat{x}_2}{2} \in C$ by the convexity of C ,
so

$$\|\mathbf{x} - \hat{x}_c\| \geq \text{dist}(\mathbf{x}, C)$$

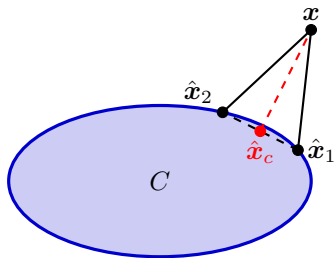
- By the parallelogram law,

$$\|\mathbf{y} + \mathbf{z}\|^2 + \|\mathbf{y} - \mathbf{z}\|^2 = 2\|\mathbf{y}\|^2 + 2\|\mathbf{z}\|^2$$

we obtain

$$\begin{aligned}\|\hat{x}_1 - \hat{x}_2\|^2 &= 2\|\mathbf{x} - \hat{x}_1\|^2 + 2\|\mathbf{x} - \hat{x}_2\|^2 - \|2\mathbf{x} - (\hat{x}_1 + \hat{x}_2)\|^2 \\ &= 4[\text{dist}(\mathbf{x}, C)]^2 - 4\|\mathbf{x} - \hat{x}_c\|^2 \leq 0\end{aligned}$$

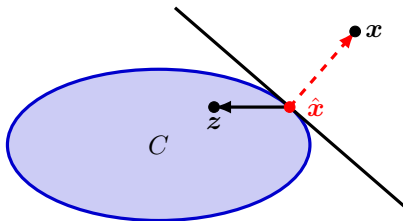
so $\hat{x}_1 = \hat{x}_2$.



Projection onto convex set (cont'd)

Proposition. Let $C \subset \mathbb{R}^n$ be nonempty, closed and convex. Given $\hat{x} \in C$, $\hat{x} = \mathcal{P}_C(x)$ iff

$$\langle x - \hat{x}, z - \hat{x} \rangle \leq 0, \quad \forall z \in C$$



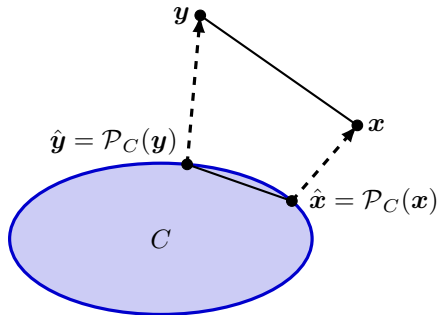
Proof. Note $\hat{x} + t(z - \hat{x}) \in C$, $\forall z \in C, t \in [0, 1]$.

$$\begin{aligned} \hat{x} = \mathcal{P}_C(x) &\iff \|x - \hat{x} - t(z - \hat{x})\|^2 \geq \|x - \hat{x}\|^2, \quad \forall z \in C, t \in [0, 1] \\ &\iff -2t\langle x - \hat{x}, z - \hat{x} \rangle + t^2\|z - \hat{x}\|^2 \geq 0, \quad \forall z \in C, t \in [0, 1] \\ &\iff \langle x - \hat{x}, z - \hat{x} \rangle \leq 0, \quad \forall z \in C \end{aligned}$$

Projection onto convex set (cont'd)

Corollary. The projection operator is nonexpansive, i.e.

$$\| \mathcal{P}_C(\mathbf{x}) - \mathcal{P}_C(\mathbf{y}) \| \leq \| \mathbf{x} - \mathbf{y} \|.$$



Proof. Let $\hat{\mathbf{x}} = \mathcal{P}_C(\mathbf{x})$, $\hat{\mathbf{y}} = \mathcal{P}_C(\mathbf{y})$. By the proposition on slide 30,

$$\begin{aligned} \| \mathbf{x} - \mathbf{y} \|^2 &= \| \hat{\mathbf{x}} - \hat{\mathbf{y}} \|^2 + \| \mathbf{x} - \mathbf{y} - (\hat{\mathbf{x}} - \hat{\mathbf{y}}) \|^2 + 2 \langle \mathbf{x} - \mathbf{y} - (\hat{\mathbf{x}} - \hat{\mathbf{y}}), \hat{\mathbf{x}} - \hat{\mathbf{y}} \rangle \\ &\geq \| \hat{\mathbf{x}} - \hat{\mathbf{y}} \|^2 - 2 \underbrace{\langle \mathbf{y} - \hat{\mathbf{y}}, \hat{\mathbf{x}} - \hat{\mathbf{y}} \rangle}_{\leq 0} - 2 \underbrace{\langle \mathbf{x} - \hat{\mathbf{x}}, \hat{\mathbf{y}} - \hat{\mathbf{x}} \rangle}_{\leq 0} \geq \| \hat{\mathbf{x}} - \hat{\mathbf{y}} \|^2 \end{aligned}$$

Projection onto convex set (cont'd)

Corollary. Let $C \subset \mathbb{R}^n$ be nonempty, closed and convex. For $x_0 \notin C$, there exists a $w \in \mathbb{R}^n \setminus \{0\}$ s.t.

$$\sup_{x \in C} \langle w, x \rangle < \langle w, x_0 \rangle.$$

Note. Special case of the separating hyperplane theorem on slide 36.

Proof. Let $\hat{x}_0 = \mathcal{P}_C(x_0)$ and $w = x_0 - \hat{x}_0$. Since $x_0 \notin C$ and $\hat{x}_0 \in C$, $w \neq 0$. By the proposition on slide 30, for any $x \in C$,

$$\langle w, x - \hat{x}_0 \rangle \leq 0$$

so

$$\langle w, x \rangle \leq \langle w, \hat{x}_0 \rangle = \langle w, x_0 \rangle - \langle w, w \rangle$$

Taking supremum over C ,

$$\sup_{x \in C} \langle w, x \rangle \leq \langle w, x_0 \rangle - \langle w, w \rangle < \langle w, x_0 \rangle$$

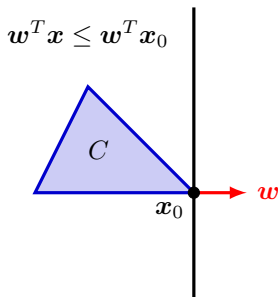
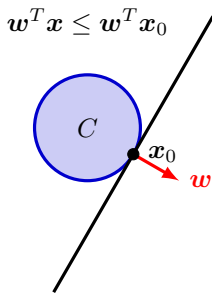
Supporting hyperplane

The **boundary** of a set C is $\partial C = \overline{C} \setminus \text{int } C$.

Supporting hyperplane theorem. If C is a nonempty, convex set in $\mathbb{R}^n$ and $x_0 \in \partial C$, then there exists $w \in \mathbb{R}^n \setminus \{0\}$ s.t.

$$\langle w, x \rangle \leq \langle w, x_0 \rangle, \quad \forall x \in C$$

$P = \{x : w^T x = w^T x_0\}$ is called a **supporting hyperplane** to C at x_0 .



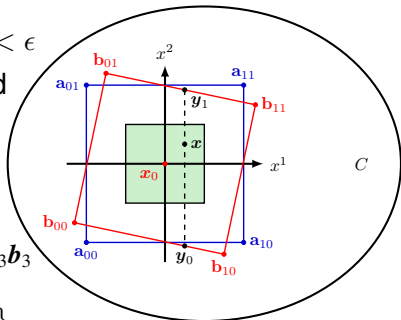
Supporting hyperplane (cont'd)

Lemma. If C is convex, then $\text{int } C = \text{int } \overline{C}$ and $\partial C = \partial \overline{C}$.

Proof sketch for 2D. First show $\text{int } C = \text{int } \overline{C}$. Let $x_0 \in \text{int } \overline{C}$. Show $x_0 \in \text{int } C$. WLOG, assume $x_0 = \mathbf{0}$. Let $\bar{B}(r) = \{x : \|x\|_\infty \leq r\}$.

1. $\exists \epsilon > 0$ s.t. $\bar{B}(2\epsilon) \subset \overline{C}$
2. Let a_i denote the vertices of $\bar{B}(2\epsilon)$
3. $a_i \in \overline{C} \implies \exists b_i \in C$ s.t. $\|a_i - b_i\|_\infty < \epsilon$
4. Show $\bar{B}(\epsilon) \subset \text{conv}\{b_0, b_1, b_2, b_3\}$ and hence $x_0 \in \text{int } C$

- ▶ Let $x \in \bar{B}(\epsilon)$
- ▶ Find θ_i s.t.
$$x^1 = \theta_0 b_0^1 + \theta_2 b_2^1 = \theta_1 b_1^1 + \theta_3 b_3^1$$
- ▶ Let $y_0 = \theta_0 b_0 + \theta_2 b_2$, $y_1 = \theta_1 b_1 + \theta_3 b_3$
- ▶ Find α s.t. $x^2 = \alpha y_0^2 + \bar{\alpha} y_1^2$
- ▶ $x = \alpha y_0 + \bar{\alpha} y_1 \in \text{conv}\{b_0, b_1, b_2, b_3\}$



Then $\partial C = \overline{C} \setminus \text{int } C = \overline{C} \setminus \text{int } \overline{C} = \partial \overline{C}$.

Supporting hyperplane (cont'd)

Proof of theorem.

- $\mathbf{x}_0 \in \partial C \implies \mathbf{x}_0 \in \partial \overline{C}$ by the previous lemma.
- There exists a sequence $\{\mathbf{x}_k\}$ s.t. $\mathbf{x}_k \notin \overline{C}$ and $\mathbf{x}_k \rightarrow \mathbf{x}_0$ as $k \rightarrow \infty$.
- By the corollary on slide 32, there exists $\mathbf{w}_k \neq 0$ s.t.

$$\langle \mathbf{w}_k, \mathbf{x} \rangle < \langle \mathbf{w}_k, \mathbf{x}_k \rangle, \quad \forall \mathbf{x} \in C$$

By rescaling, we can assume $\|\mathbf{w}_k\| = 1$.

- By the Bolzano-Weierstrass Theorem, $\{\mathbf{w}_k\}$ has a convergent subsequence $\mathbf{w}_{k_i} \rightarrow \mathbf{w}$.
- Taking the limit $i \rightarrow \infty$ along the subsequence,

$$\langle \mathbf{w}_{k_i}, \mathbf{x} \rangle < \langle \mathbf{w}_{k_i}, \mathbf{x}_{k_i} \rangle, \quad \forall \mathbf{x} \in C \implies \langle \mathbf{w}, \mathbf{x} \rangle \leq \langle \mathbf{w}, \mathbf{x}_0 \rangle, \quad \forall \mathbf{x} \in C$$

$$\|\mathbf{w}_{k_i}\| = 1 \implies \|\mathbf{w}\| = 1$$

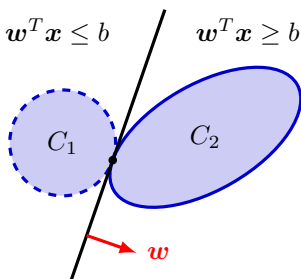
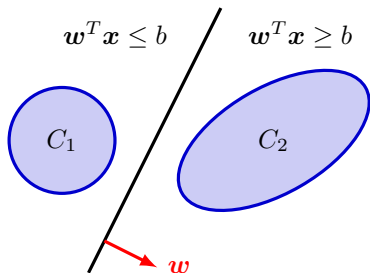
Separating hyperplane

Separating hyperplane theorem. If C_1, C_2 are nonempty, convex sets in $\mathbb{R}^n$ with $C_1 \cap C_2 = \emptyset$, then C_1 and C_2 can be separated by a hyperplane, i.e. there exists $w \in \mathbb{R}^n \setminus \{0\}$, $b \in \mathbb{R}$ s.t.

$$w^T x \leq b, \quad \forall x \in C_1$$

$$w^T x \geq b, \quad \forall x \in C_2$$

$P = \{x : w^T x = b\}$ is called a **separating hyperplane** for C_1 and C_2 .



Separating hyperplane (cont'd)

Lemma. If C_1, C_2 are two nonempty convex sets s.t. $C_1 \cap C_2 = \emptyset$, then $C = C_1 - C_2 = \{\mathbf{x}_1 - \mathbf{x}_2 : \mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2\}$ is a nonempty convex set and $\mathbf{0} \notin C$.

Proof of theorem.

- It suffices to show there exists a $\mathbf{w} \neq \mathbf{0}$ s.t.

$$\mathbf{w}^T \mathbf{x}_1 \leq \mathbf{w}^T \mathbf{x}_2, \quad \forall \mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2. \quad (\dagger)$$

Then we can take $b = \sup_{\mathbf{x}_1 \in C_1} \mathbf{w}^T \mathbf{x}_1 \in \mathbb{R}$.

- Let $C = C_1 - C_2$. Then $(\dagger)$ is equivalent to

$$\langle \mathbf{w}, \mathbf{x} \rangle \leq 0, \quad \forall \mathbf{x} \in C. \quad (\ddagger)$$

- Since $\mathbf{0} \notin C$, there are two cases.
 - ▶ If $\mathbf{0} \notin \overline{C}$, $(\ddagger)$ will follow from the corollary on slide 32.
 - ▶ If $\mathbf{0} \in \partial C$, $(\ddagger)$ will follow from the supporting hyperplane theorem.